OPEN-SOURCE TECHNOLOGIES
Average Assignment

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Class: 6EC2
Batch: B

1. **Git Workflow Practice:**

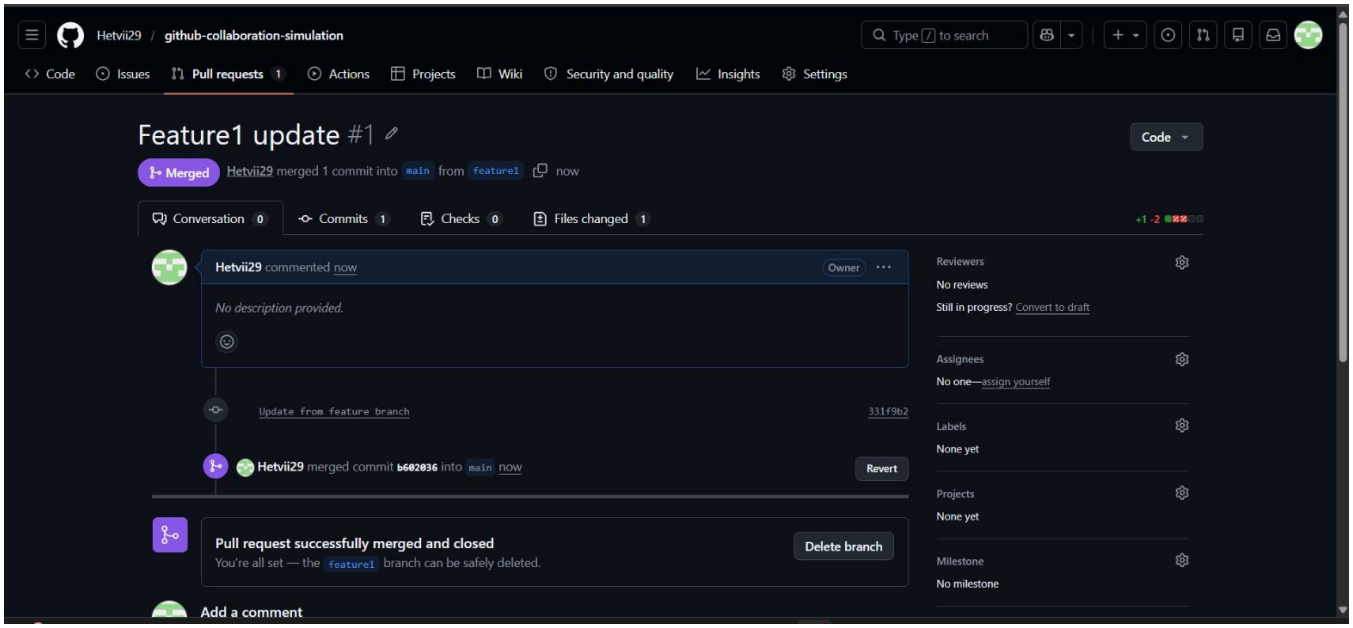
```
C:\Users\hp>cd OST_Lab

C:\Users\hp\OST_lab>mkdir p1

C:\Users\hp\OST_lab>cd p1

C:\Users\hp\OST_lab\p1>git init
Initialized empty Git repository in C:/Users/hp/OST_lab/p1/.git/

C:\Users\hp\OST_lab\p1>git status
On branch master
No commits yet
nothing to commit (create/copy files and use "git add" to track)

C:\Users\hp\OST_lab\p1>git  add  .

C:\Users\hp\OST_lab\p1>git commit -m "Initial Commit"
[master (root-commit) 65214d1] Initial Commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 1.html

C:\Users\hp\OST_lab\p1>git  branch
*  master

C:\Users\hp\OST_lab\p1>git  branch  feature1

C:\Users\hp\OST_lab\p1>git checkout feature1
Switched to branch 'feature1'

C:\Users\hp\OST_lab\p1>git  branch
* feature1
  master

C:\Users\hp\OST_lab\p1>git status
On branch feature1
nothing to commit, working tree clean

C:\Users\hp\OST_lab\p1>git add .

C:\Users\hp\OST_lab\p1>
```

```
C:\Users\hp>cd OST_Lab

C:\Users\hp\OST_lab>mkdir p1

C:\Users\hp\OST_lab>cd p1

C:\Users\hp\OST_lab\p1>git init
Initialized empty Git repository in C:/Users/hp/OST_lab/p1/.git/

C:\Users\hp\OST_lab\p1>git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)

C:\Users\hp\OST_lab\p1>git add .

C:\Users\hp\OST_lab\p1>git commit -m "Initial Commit"
[master (root-commit) 65214d1] Initial Commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 1.html

C:\Users\hp\OST_lab\p1>git branch
* master

C:\Users\hp\OST_lab\p1>git branch feature1

C:\Users\hp\OST_lab\p1>git checkout feature1
Switched to branch 'feature1'

C:\Users\hp\OST_lab\p1>git branch
* feature1
  master

C:\Users\hp\OST_lab\p1>git status
On branch feature1
nothing to commit, working tree clean

C:\Users\hp\OST_lab\p1>git add .

C:\Users\hp\OST_lab\p1>
```

A **branch** in Git is a separate version of the project where changes can be made without affecting the main code.
The feature1 branch allows us to work on new features safely.
Once changes are complete, it can be merged back into the main branch.

2. GitHub Collaboration Simulation

Link: https://github.com/Hetvii29



3. **Use Package Manager**

With use of npm:

```
C:\Users\hp\OST_lab>mkdir p3

C:\Users\hp\OST_lab>cd p3

C:\Users\hp\OST_lab\p3>npm init -y
Wrote to C:\Users\hp\OST_lab\p3\package.json:

{
  "name": "p3",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" fifi exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}

C:\Users\hp\OST_lab\p3>npm install express
added 65 packages, and audited 66 packages in 7s
22 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities

C:\Users\hp\OST_lab\p3>npm install axios
added 11 packages, and audited 77 packages in 5s
24 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities

C:\Users\hp\OST_lab\p3>npm list
p3@1.0.0 C:\Users\hp\OST_lab\p3
+-- axios@1.14.0
`-- express@5.2.1


C:\Users\hp\OST_lab\p3>
```

With use of pip:

```
C:\Users\hp\OST_lab\p3>pip install numpy
Requirement already satisfied: numpy in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(2.1.2)
[notice] A new release of pip available: 22.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\hp\OST_lab\p3>pip install requests
Requirement already satisfied: requests in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(2.32.3)
Requirement already satisfied: charset-normalizer<4,>=2 in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(from requests) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(from requests) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(from requests) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(from requests) (2026.2.25)
[notice] A new release of pip available: 22.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\hp\OST_lab\p3>pip list | findstr "numpy requests"
numpy                    2.1.2
requests                 2.32.3

[notice] A new release of pip available: 22.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\hp\OST_lab\p3>
```

```
C:\Users\hp\OST_lab>mkdir p3

C:\Users\hp\OST_lab>cd p3

C:\Users\hp\OST_lab\p3>npm init -y
Wrote to C:\Users\hp\OST_lab\p3\package.json:

{
  "name": "p3",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}




C:\Users\hp\OST_lab\p3>npm install express

added 65 packages, and audited 66 packages in 7s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\hp\OST_lab\p3>npm install axios

added 11 packages, and audited 77 packages in 5s

24 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\hp\OST_lab\p3>npm list
p3@1.0.0 C:\Users\hp\OST_lab\p3
+-- axios@1.14.0
`-- express@5.2.1


C:\Users\hp\OST_lab\p3>
```

```
C:\Users\hp\OST_lab\p3>pip install numpy
Requirement already satisfied: numpy in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (2.1.2)

[notice] A new release of pip available: 22.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\hp\OST_lab\p3>pip install requests
Requirement already satisfied: requests in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (2.32.3)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (from requests) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (from requests) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (from requests) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (from requests) (2026.2.25)

[notice] A new release of pip available: 22.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\hp\OST_lab\p3>pip list | findstr "numpy requests"
numpy                     2.1.2
requests                  2.32.3

[notice] A new release of pip available: 22.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\hp\OST_lab\p3>
```

4. Semantic Versioning

For npm:

```
C:\Users\hp\OST_lab>mkdir p4

C:\Users\hp\OST_lab>cd p4

C:\Users\hp\OST_lab\p4>npm init -y
Wrote to C:\Users\hp\OST_lab\p4\package.json:
{
  "name": "p4",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" fifi exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}

C:\Users\hp\OST_lab\p4>npm install lodash
added 1 package, and audited 2 packages in 4s
found 0 vulnerabilities
C:\Users\hp\OST_lab\p4>npm list lodash
p4@1.0.0 C:\Users\hp\OST_lab\p4
`-- lodash@4.18.1

C:\Users\hp\OST_lab\p4>npm install lodash @4.17.20
npm error code EINVALIDTAGNAME
npm error Invalid tag name "@4.17.20" of package "@4.17.20": Tags may
not have any characters that encodeURIComponent encodes.
npm error A complete log of this run can be found in:
C:\Users\hp\AppData\Local\npm-cache\_logs\2026-04-04T10_45_58_312Z-
debug-0.log

C:\Users\hp\OST_lab\p4>npm install lodash@4.17.20
changed 1 package, and audited 2 packages in 2s
1 high severity vulnerability
To address all issues, run:
  npm audit fix
Run `npm audit` for details.

C:\Users\hp\OST_lab\p4>npm list lodash
p4@1.0.0 C:\Users\hp\OST_lab\p4
`-- lodash@4.17.20

C:\Users\hp\OST_lab\p4>
```

For pip:

```
C:\Users\hp\OST_lab\p4>pip install requests
Requirement already satisfied: requests in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(2.32.3)
Requirement already satisfied: charset-normalizer<4,>=2 in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(from requests) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(from requests) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(from requests) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(from requests) (2026.2.25)

[notice] A new release of pip available: 22.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\hp\OST_lab\p4>pip show requests
Name: requests
Version: 2.32.3
Summary: flython HTTfl for Humans.
Home-page: https://requests.readthedocs.io
Author: Kenneth Reitz
Author-email: me@kennethreitz.org
License: Apache-2.0
Location:
C:\Users\hp\AppData\Local\flrograms\flython\flython311\Lib\site-packages
Requires: certifi, charset-normalizer, idna, urllib3
Required-by: jupyterlab_server, webdriver-manager

C:\Users\hp\OST_lab\p4>pip install requests==2.28.1
Collecting requests==2.28.1
  Downloading requests-2.28.1-py3-none-any.whl (62 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 62.8/62.8 kB 420.8 kB/s
eta 0:00:00
Collecting charset-normalizer<3,>=2
  Downloading charset_normalizer-2.1.1-py3-none-any.whl (39 kB)
Requirement already satisfied: idna<4,>=2.5 in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(from requests==2.28.1) (3.7)
Collecting urllib3<1.27,>=1.21.1
  Downloading urllib3-1.26.20-py2.py3-none-any.whl (144 kB)
```

```
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 144.2/144.2 kB 1.1 MB/s
eta 0:00:00
Requirement already satisfied: certifi>=2017.4.17 in
c:\users\hp\appdata\local\programs\python\python311\lib\site-packages
(from requests==2.28.1) (2026.2.25)
Installing collected packages: urllib3, charset-normalizer, requests
  Attempting uninstall: urllib3
    Found existing installation: urllib3 2.6.3
    Uninstalling urllib3-2.6.3:
      Successfully uninstalled urllib3-2.6.3
  Attempting uninstall: charset-normalizer
    Found existing installation: charset-normalizer 3.3.2
    Uninstalling charset-normalizer-3.3.2:
      Successfully uninstalled charset-normalizer-3.3.2
  Attempting uninstall: requests
    Found existing installation: requests 2.32.3
    Uninstalling requests-2.32.3:
      Successfully uninstalled requests-2.32.3
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
jupyterlab-server 2.27.3 requires requests>=2.31, but you have
requests 2.28.1 which is incompatible.
selenium 4.41.0 requires urllib3[socks]<3.0,>=2.6.3, but you have
urllib3 1.26.20 which is incompatible.
Successfully installed charset-normalizer-2.1.1 requests-2.28.1
urllib3-1.26.20

[notice] A new release of pip available: 22.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\hp\OST_lab\p4>pip show requests
Name: requests
Version: 2.28.1
Summary: flython HTTfl for Humans.
Home-page: https://requests.readthedocs.io
Author: Kenneth Reitz
Author-email: me@kennethreitz.org
License: Apache 2.0
Location:
C:\Users\hp\AppData\Local\flrograms\flython\flython311\Lib\site-packages
Requires: certifi, charset-normalizer, idna, urllib3
Required-by: jupyterlab_server, webdriver-manager

C:\Users\hp\OST_lab\p4>
```

```
C:\Users\hp\OST_lab>mkdir p4

C:\Users\hp\OST_lab>cd p4

C:\Users\hp\OST_lab\p4>npm init -y
Wrote to C:\Users\hp\OST_lab\p4\package.json:

{
  "name": "p4",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}



C:\Users\hp\OST_lab\p4>npm install lodash

added 1 package, and audited 2 packages in 4s

found 0 vulnerabilities

C:\Users\hp\OST_lab\p4>npm list lodash
p4@1.0.0 C:\Users\hp\OST_lab\p4
`-- lodash@4.18.1


C:\Users\hp\OST_lab\p4>npm install lodash @4.17.20
npm error code EINVALIDTAGNAME
npm error Invalid tag name "@4.17.20" of package "@4.17.20": Tags may not have any characters that encodeURIComponent encodes.
npm error A complete log of this run can be found in: C:\Users\hp\AppData\Local\npm-cache\_logs\2026-04-04T10_45_58_312Z-debug-0.log

C:\Users\hp\OST_lab\p4>npm install lodash@4.17.20

changed 1 package, and audited 2 packages in 2s

1 high severity vulnerability

To address all issues, run:
  npm audit fix

Run `npm audit` for details.

C:\Users\hp\OST_lab\p4>npm list lodash
p4@1.0.0 C:\Users\hp\OST_lab\p4
`-- lodash@4.17.20
```

```
C:\Users\hp\OST_lab\p4>pip install requests
Requirement already satisfied: requests in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (2.32.3)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (from requests) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (from requests) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (from requests) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (from requests) (2026.2.25)

[notice] A new release of pip available: 22.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\hp\OST_lab\p4>pip show requests
Name: requests
Version: 2.32.3
Summary: Python HTTP for Humans.
Home-page: https://requests.readthedocs.io
Author: Kenneth Reitz
Author-email: me@kennethreitz.org
License: Apache-2.0
Location: C:\Users\hp\AppData\Local\Programs\Python\Python311\Lib\site-packages
Requires: certifi, charset-normalizer, idna, urllib3
Required-by: jupyterlab_server, webdriver-manager

C:\Users\hp\OST_lab\p4>pip install requests==2.28.1
Collecting requests==2.28.1
  Downloading requests-2.28.1-py3-none-any.whl (62 kB)
     ---------------------------------------- 62.8/62.8 kB 420.8 kB/s eta 0:00:00
Collecting charset-normalizer<3,>=2
  Downloading charset_normalizer-2.1.1-py3-none-any.whl (39 kB)
Requirement already satisfied: idna<4,>=2.5 in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (from requests==2.28.1) (3.7)
Collecting urllib3<1.27,>=1.21.1
  Downloading urllib3-1.26.20-py2.py3-none-any.whl (144 kB)
     ---------------------------------------- 144.2/144.2 kB 1.1 MB/s eta 0:00:00
Requirement already satisfied: certifi>=2017.4.17 in c:\users\hp\appdata\local\programs\python\python311\lib\site-packages (from requests==2.28.1) (2026.2.25)
Installing collected packages: urllib3, charset-normalizer, requests
  Attempting uninstall: urllib3
    Found existing installation: urllib3 2.6.3
    Uninstalling urllib3-2.6.3:
      Successfully uninstalled urllib3-2.6.3
  Attempting uninstall: charset-normalizer
    Found existing installation: charset-normalizer 3.3.2
    Uninstalling charset-normalizer-3.3.2:
      Successfully uninstalled charset-normalizer-3.3.2
  Attempting uninstall: requests
    Found existing installation: requests 2.32.3
    Uninstalling requests-2.32.3:
      Successfully uninstalled requests-2.32.3
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
jupyterlab-server 2.27.3 requires requests>=2.31, but you have requests 2.28.1 which is incompatible.
selenium 4.41.0 requires urllib3[socks]<3.0,>=2.6.3, but you have urllib3 1.26.20 which is incompatible.
Successfully installed charset-normalizer-2.1.1 requests-2.28.1 urllib3-1.26.20

[notice] A new release of pip available: 22.3 -> 26.0.1
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C:\Users\hp\OST_lab\p4>pip show requests
Name: requests
Version: 2.28.1
Summary: Python HTTP for Humans.
Home-page: https://requests.readthedocs.io
Author: Kenneth Reitz
Author-email: me@kennethreitz.org
License: Apache 2.0
Location: C:\Users\hp\AppData\Local\Programs\Python\Python311\Lib\site-packages
Requires: certifi, charset-normalizer, idna, urllib3
Required-by: jupyterlab_server, webdriver-manager

C:\Users\hp\OST_lab\p4>
```